



Seeing Right Through You

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Field(s) of Interest: Physics, Biology, Medicine

Brief Overview (1-3 sentences):

In this lesson, mentees will learn about how X-Ray machines work, what X-Rays are, and the various ways light can interact with matter.

Agenda:

- Introduction (5 min)
- Module 1: Big Ah Machine (20 min)
- Module 2: Armour up! (20 min)
- Module 3: Ouch my Bone! (10 min)
- Conclusion (5 min)

Main Teaching Goals/Key Terms:

→ **Waves**

◆ **X-rays**

→ **Transmission**

→ **Absorption**

→ **X-Ray Machine**

→ **Density**

→ **Lead Vest**

→ **Ionizing Radiation**

→ **Non-Visible Light**

Background for Mentors

Module 1

- Waves
 - X-rays
- Transmission
- Absorption
- X-Ray Machine

Waves in the context of physics are *disturbances that transfer energy from one place to another through matter or a medium*. In this lesson, we are focused on a specific type of wave called Electromagnetic waves. Electromagnetic waves do not require a medium to travel and include visible lights and X-rays.

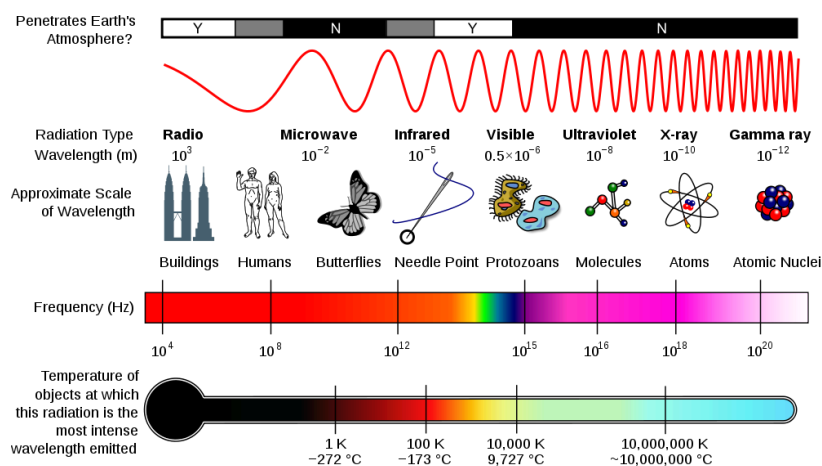


Figure 1. The various forms of waves

X-Rays are a *type of electromagnetic wave that do not require a medium to travel through*. They are high energy and high frequency which allows them to penetrate through soft, low density materials such as skin and tissue, but are absorbed by higher density materials such as bones. Because they carry high amounts of energy, they have the energy to ionize electrons from atoms, or in other words, remove electrons from atoms.

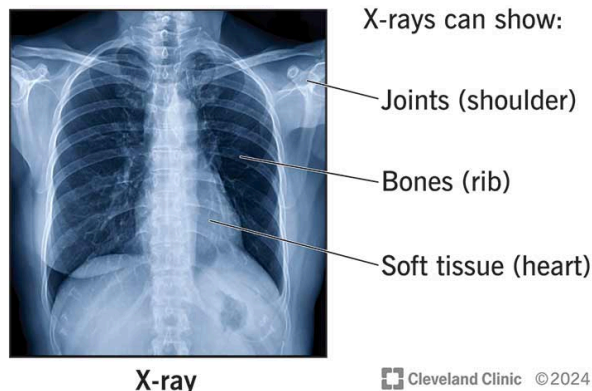


Figure 2. An X-Ray image of the body

Transmission is the process of waves passing through a material or medium without being absorbed or reflected. X-Rays are able to transmit through soft, low density material such as skin and tissue.

Absorption is the process where matter absorbs the energy from a wave and turns it into internal energy. When an X-Ray encounters a hard, dense material such as bone, it does not have enough energy to transmit through and therefore is absorbed by the bone. The energy absorbed by the bone is transferred into ionization energy in which the X-ray causes electrons in the atoms of your bone to be emitted (photoelectric effect).

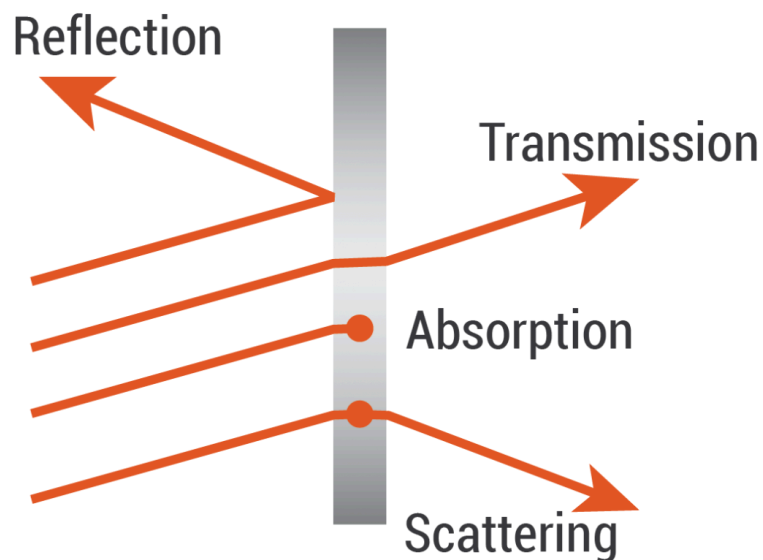


Figure 3. An image showing the various ways a wave interacts with matter

An **X-ray machine** can be separated into two parts:

1) producing X-rays 2) detecting the amount of X-rays that are transmitted.

- 1) X-rays are produced through an X-ray tube. The tube consists of a cathode (-) tungsten filament which is heated to the point where electrons are emitted from the surface through the process of thermionic emission. Electrons are accelerated through a high potential difference between the cathode and anode. The electrons then collide with the tungsten anode (+) to produce X-rays.
- 2) Detectors are placed on the far side of the patient and pick up on the transmitted X-Rays. Essentially it maps out how much each X-ray passed through the parts of your body.

X-ray tube

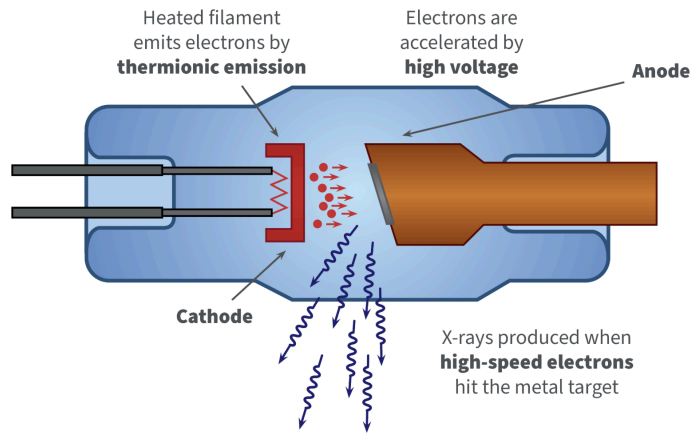


Figure 4. An image of an X-ray tube

X-ray

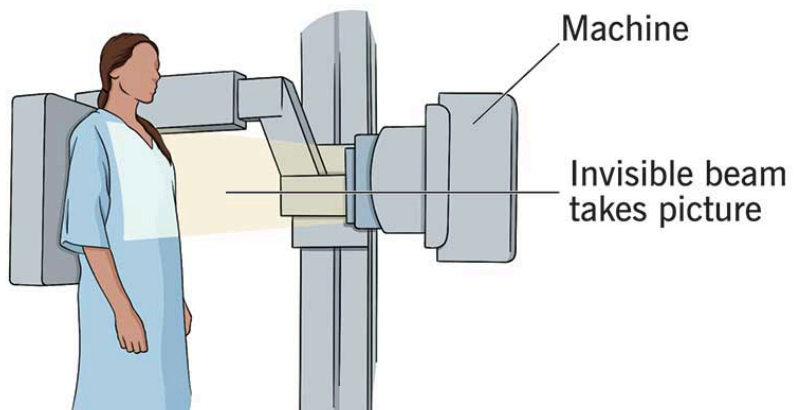


Figure 5. An image of an X-ray Machine

Module 2

- Density
- Transmission
- Absorption
- Lead Vest
- Ionizing Radiation

Density is defined as *the amount of mass per volume*. In other words, it is the measure of how much stuff can fit in a given amount of space. Bones are more dense compared to tissue and skin, which is why bones absorb X-rays more because the X-Rays have a higher probability of interacting with atoms in bones compared to tissue and skin and therefore are absorbed by the bones.

Component grouping	Mass density (kg m ⁻³)	Young's modulus (MPa)
<i>Adipose</i>	950 ^a	Evaluated
<i>Bone</i>	1920 ^a	18,300 ^c
<i>Cartilage</i>	1100 ^a	25 ^c
<i>Ligament/Tendon</i>	1170 ^b	39 ^d
<i>Membrane</i>	1100	528 ^d
<i>Muscle</i>	1050 ^a	Evaluated

^a [ICRP] International Commission on Radiological Protection (2009).

^b Duck (1990).

Figure 5. A table showing the various densities that make up your body

A **Lead Vest** is a *protective garment made of lead and other various heavy metals used during X-Ray imaging to protect vital organs from **Ionizing Radiation**.*



Figure 6. An image of a Lead vest during an X-Ray image

Ionizing Radiation is radiation whose photons (X-Rays) carry enough energy to excite electrons from atoms. If these X-Rays penetrate through certain vital organs, it can potentially cause DNA damage which could lead to harmful effects.

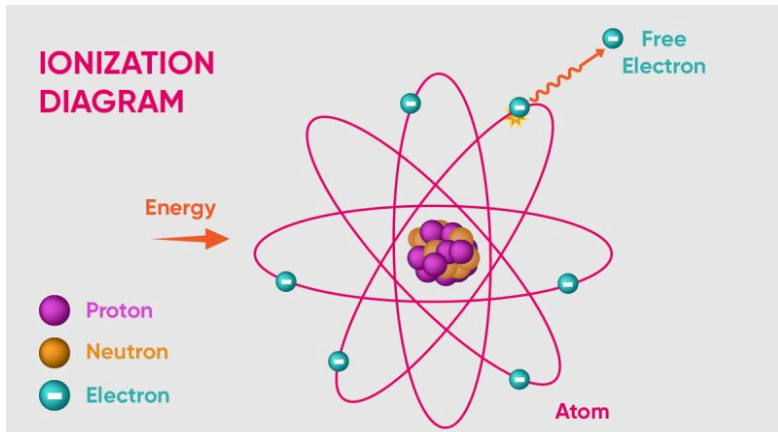


Figure 7. A diagram of ionizing radiation

Module 3

- Density
- Transmission
- Absorption
- Non-Visible Light

Non-Visible Light refers to the parts of the electromagnetic spectrum where the human eyes are unable to detect either too short or too long of wavelengths. Non-visible light includes UV, X-Rays, Gamma rays, Infrared, Radio waves, and Microwaves.

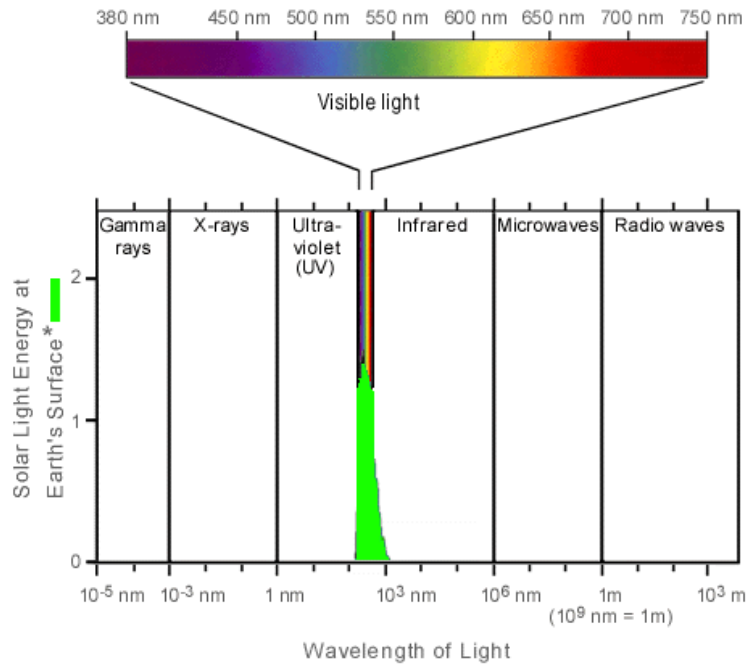


Figure 8. A diagram of the electromagnetic spectrum

Introduction

Concepts to Introduce • Electromagnetic waves travel like the waves we see in the ocean, except we can't see them. They can travel through matter or in a vacuum. • Transmission is when a wave is able to travel through matter. • Absorption is when a wave is unable to fully travel through matter and is absorbed by the matter. • Density is how much stuff you can fit in a given space. • X-Rays are a form of Electromagnetic waves that are high energy and high frequency.	Questions to Pique Interest • Have you ever broken a bone? • Have you ever wondered how your X-Ray image is taken when you get an X-Ray done at the dentist or doctors? • Why can't we see the X-Rays when we're getting an image done?
Scientists, Current and Past Events • Wilhelm Roentgen discovered X-Rays accidentally and called them "X" rays since he did not know what they were. • X-Rays are used in many applications aside from the medical field such as in airport security, inspection in planes, and studying distant space objects.	Careers and Applications • Radiologists ○ Radiologists are physicians who specialize in diagnosing and treating disease using medical imaging. They interpret X-rays, CT, MRI, ultrasound, mammography, and nuclear medicine scans. • X-Ray Technician ○ X-Ray Technicians are the healthcare professionals who actually perform the imaging exams. They position patients, set the exposure on the equipment, take the images, and check that the pictures are clear enough for the radiologist to read.

Module 1: A Big Ah Machine

This will be a group activity in which mentees are split up into groups of 4-5. The mentees will be building a model X-ray machine to learn about how an X-Ray machine works.

Teaching Goals

1. **Waves:** A movement of energy from one place to another.
 - a. **X-Rays:** An electromagnetic wave that does not need a medium to travel through one place to another.
2. **X-Ray Machine:** A medical imaging device that takes images of the inside of your body through X-rays.
3. **Transmission:** When a wave is able to travel through matter or an object without being stopped.
4. **Absorption:** When a wave is not able to travel through matter or an object fully, and the energy of the wave is transferred to the material.

Materials

- Cardboard box
- Gluedots
- Pipe cleaners
- Parchment paper
- Flashlight

Different Methods for Teaching

1. **Waves:** Have mentees think about the string moving up and down. This is also how an X-Ray moves but instead of in one place, it travels through space.
2. **Transmission:** Have mentees think about how there is still light in the room from the sunlight going through the windows. Light is able to transmit through the glass.
3. **Absorption:** Have you ever been sunburned? This is because you absorbed too much UV light. This is the same absorption that occurs in X-Rays when our body is exposed to it.

Procedure

1. Organize groups of 4-5 mentees and have them work together to build a fun shape out of pipecleaners!
2. Help each group gluedot their amazing designs onto the precut parchment paper
3. Turn off the lights
4. With one group at a time, tape the parchment paper to the large cutout of the box with the

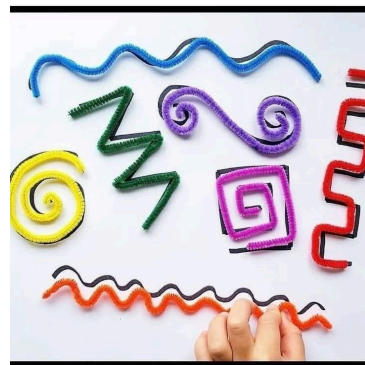


Figure 1: Fun designs the mentees can make out of the pipe cleaners

design facing inside the box.

a. Make sure the box is fully closed

5. Face the contraption towards the whiteboard or a blank wall and shine the flashlight inside the box!

6. Ask the mentees what they see and what they do not see

a. e.g. they see their design and don't see the paper

7. Emphasize the concept of Absorption and Transmission

a. e.g. their designs absorbed the light's energy and the light was able to transmit through the paper

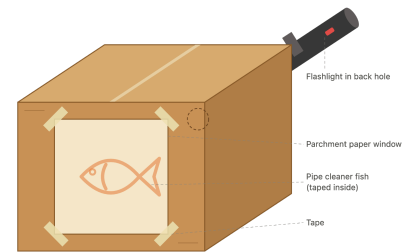


Figure 2: A crude example of what the contraption will look like

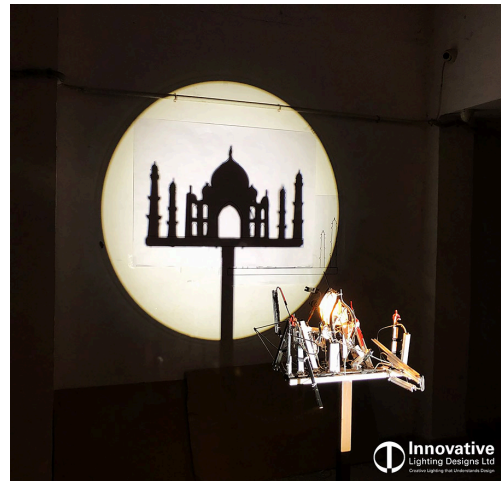


Figure 3: This is an example of what the mentees will see

Classroom Notes

Be sure that the pipe cleaners are facing inside the box and that the box is fully sealed before shining the light.

Module 2: Armor up!

Mentee will learn about why we need lead vests during an X-Ray and what these vests do to protect our organs. The beginning will be individual and at the end they will work with their classmates.

Teaching Goals

1. **Density:** The amount of mass per unit volume or in other words how packed matter is in a specified volume.
2. **Transmission:** When a wave is able to travel through matter or an object without being stopped.
3. **Absorption:** When a wave is not able to travel through matter or an object fully, and the energy of the wave is transferred to the material.
4. **Lead vest:** a protective garment made of lead and other various heavy metals used during X-Ray imaging to protect vital organs from **Ionizing Radiation**.
5. **Ionizing Radiation:** A high form of energy that travels through space that removes electrons from molecules and atoms as it passes through materials.

Materials

- Sun print paper
- Cardstock
- Markers
- Clear sheet protector

Different Methods for Teaching

1. **Lead Vest:** Have the mentees think about what a shield does. It protects the person from dangerous things. A lead vest protects our vital organs from X-Rays.
2. **Ionizing Radiation:** An example could be a ghost that is very heavy and as it passes through your body it steals some of your energy balls inside you.

Procedure

1. Pass out markers and cardstock to each mentee and have them draw their favorite animal or human or a cool design on the sheet
2. Have them carefully cut out their designed drawing and set it aside. This is the “shield”
3. With their clear protector sheet, have the mentees trace their cardstock design on the clear protector sheet with the marker. The mentees can also draw a heart or any design



Figure 1: “Sunprint paper after exposed to UV”

inside their drawn design.

4. Pass out the sunprint paper and have the mentees slide the paper into the protector sheet, being sure that the sunprint paper is directly underneath their drawing
5. Bring back the “shield” and have the mentees cover their precious heart with the “shield”
6. Bring their designs outside and place in a spot with full sunlight
7. Finish Module 3 and then check if their shields protected their designs from the harmful UV rays.

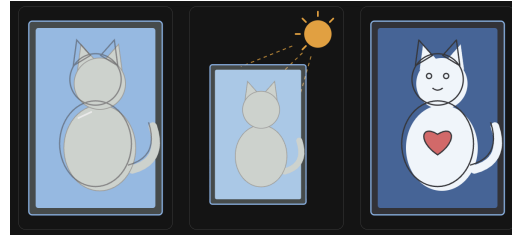


Figure 2: An example of their designs

Classroom Notes

The longer the UV exposure the better the results will be. If it is windy outside, try to find a rock or something heavy to sit on the cardstock paper so it stays in place. For younger mentees, the mentors can help them cut out their “shield” designs. If your site has a lot of mentees and it is cloudy outside, you might want to start with this module to give the sunprint paper longer time to expose to UV for better results.

Module 3: Ouch my Bone!

Mentees will learn about what X-Rays show and learn more about the concepts of transmission and absorption.

Teaching Goals 1. Transmission: When a wave is able to travel through matter or an object without being stopped. 2. Absorption: When a wave is not able to travel through matter or an object fully, and the energy of the wave is transferred to the material. 3. Nonvisible Light: Light we can not see with our eyes. The three types of nonvisible light are: UV, X-rays, and Gamma Rays 4. Density: The amount of mass per unit volume or in other words how packed matter is in a specified volume.	Materials • Clear plastic sheet protector • Black marker • White paper • Black paper
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Different Methods for Teaching 1. Density: Have mentees think about if they were an X-ray and running into a brick wall vs running into a paper wall. The brick wall is more dense than the paper wall, so they are able to transmit through the paper wall but are absorbed by the brick wall. 2. Nonvisible Light: For older mentees you can talk about how visible light is only visible because our eyes can only see 380nm to 750nm wavelengths. Anything below or above these limits, our eyes can not see. For younger mentees, you can tell them that there are many waves when a microwave is heating our food, but we can not see these waves.

Procedure 1. Pass out clear protector sheets and black markers to each of the Mentees. 2. Have them draw an outline of a human body. a. You can have of the Mentors draw an example of what to draw on the whiteboard 3. After the human body outline is drawn, have the Mentees draw in bones inside the body	
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with one part of the bone broken.

- a. Mentees can also draw other things inside the body. The point of the exercise is for them to find something with their “light”
4. Pass out the black sheets of paper and the white paper “the detector” to each Mentee and have them insert the black paper into the clear protector sheets.
5. Have the Mentees swap their projects with their friends or classmates.
6. With their white piece of paper, have them insert the paper in between the black paper and the clear protector sheet and have them scan the body for the broken bone or whatever they drew!



Figure 1: Clear protector sheet and Black paper

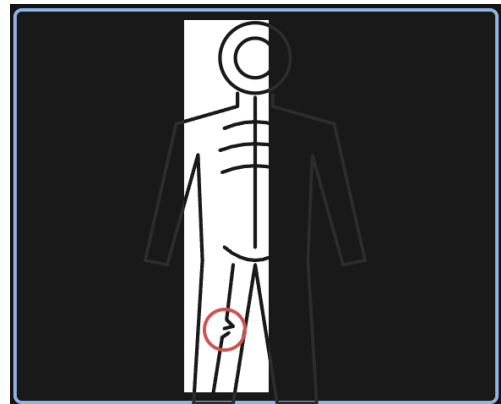


Figure 2: The white paper scanned for the broken bone!

Classroom Notes

Make sure the mentees do not accidentally erase their marker drawings.

Conclusion

Ask the mentees if next time they get an X-Ray done if they can explain to their doctor how it works!

Summary Materials Table

Material	Amount per Site	Expected \$\$	Vendor (or online link)
Cardboard Box	1 per 4-5 mentees	Free	USPS
Gluedots	2 per mentee	In HC	
Pipecleaners	1 per mentee	In HC	
Parchment Paper	1 per 4-5 mentees	16\$	Amazon
Flashlights	1 per box	16\$	Amazon
Sun Print Paper	1 per mentee	23\$	Amazon
Cardstock	1 per 2-3 mentees	16\$	Amazon
Markers	Good amount per site	In HC	
Clear Protector Sheets	2 per mentee	18\$	Amazon
White Paper	1 per mentee	In HC	
Black Paper	1 per mentee	In HC	
Scissors	Good amount per site	In HC	